

How each number is calculated

Short definitions for every screen, heading, and card

Read this first

This document explains what each heading and card means, how the number is built, and which live system it comes from. Overview is covered first and in most detail because that is the chain summary. Branch, Reports, Statistics, and Schedules follow.

Three data systems

- Home Assistant (HA): live device states every few seconds. Power, voltage, climate, lights, cameras, automations.
- Energy SQLite (data/energy.sqlite): one 6:00 AM Asia/Riyadh snapshot of each kWh meter. Used for closed days, month-to-date, and comparisons.
- LC Logic (visitor cameras): people counts, table zones, entries, hourly and daily visits. Customer visits always mean entries, not camera events.

Energy day (used everywhere energy is summed)

An energy day does not follow midnight. It starts at 6:00 AM Asia/Riyadh. Before 6:00 AM, readings still belong to yesterday. Cron writes one snapshot of every meter at 6:00 AM. Closed-day kWh = next morning snapshot minus this morning snapshot. Today so far (live) = current HA meter minus today's 6:00 AM snapshot. Negative values are clamped to 0. Meters never reset in Home Assistant.

Important split on Overview

- Energy month-to-date, week, year, charts, and "Energy today" on Overview use CLOSED 6:00 AM days only. Today's open day is not in those totals yet.
- Visitor month-to-date and "Visitors today" include live LC Logic entries for the calendar day (Riyadh).
- Live today kWh (since 6:00 AM) is on Branch and Reports, not in the Overview KPI totals.
- Cost uses a fixed tariff of 0.20 SAR per kWh (SEC mockup rate in the app).
- Overview is admin-only. Employees land on Branch.

1. Overview (admin)

Route: /. Header clock is Asia/Riyadh, labelled Al Malqa, Riyadh. Range pills (Today / Month to date / Quarter / Year) change the first energy, cost, visitors, kWh per customer, and saved cards. They do not change Visitors today so far, Average dwell time, or Table use now.

Header

Clock | Al Malqa, Riyadh

Browser time formatted in Asia/Riyadh. Location is hardcoded to this live branch.

Source: Device clock. Not from HA or SQLite.

Range pills: Today / Month to date / Quarter / Year

Selects which closed-day energy sum and which visitor sum feed the KPI grid. Default is Month to date.

Source: UI only. Energy sums from SQLite closed days. Visitor sums from LC Logic.

Needs attention

Heading: Needs attention

Exception cards only, max 6. Empty state: "No open exceptions right now." Click a card to open Branch.

Source: Home Assistant states + LC Logic occupancy + whether SQLite has any closed day.

Cooling

A climate unit is on and room temperature is at least 6 C above the setpoint.

Formula: *Show if (now - target) >= 6 and state is not off.*

Source: HA climate.* | current_temperature minus temperature (target).

Waste

No one on the floor, no dining tables occupied, but AC is still drawing power.

Formula: *Show if occupiedTables = 0 and occupancy = 0 and AC power > 80 W.*

Source: LC Logic occupancy and table zones + HA sensor.ac_energy_monitor_energy1_power (fallback: energy total).

Data

No closed energy day has been written yet, so Overview charts have nothing real to sum.

Formula: *Show if closedThrough is empty.*

Source: SQLite snapshots table.

Power

Lighting circuit voltages disagree by 40 V or more.

Formula: *Show if max(voltage) - min(voltage) >= 40.*

Source: HA voltage sensors on counter, kitchen, office, oven, and dining light breakers.

Table occupancy chips (under Needs attention)

Table / zone chips

One chip per camera zone except entrance. Green = at least one person in that zone right now. Grey = empty.

Formula: *Occupied if zone count > 0.*

Source: LC Logic GET /api/counts. Same chips as Branch.

KPI cards

Energy today / month to date / this quarter / this year

Sum of Lights + AC + Oven + Freezer + Chiller kWh for the selected range. Today = last CLOSED energy day (not live since 6 AM). MTD = 1st of this month through last closed day. Quarter / year = matching months from the 12-month series, still closed days only.

Formula: *kWh = sum of device (snapshot[day+1] - snapshot[day]).*

Source: SQLite snapshots, daily consumption = next 6 AM meter minus this 6 AM meter.

Energy deltas (MTD only)

Percent vs last month, same number of closed days. Percent vs the same month last year, same days.

Formula: $\text{change \%} = (\text{current} - \text{previous}) / \text{previous} * 100$.

Source: SQLite.

Energy cost (today / MTD / quarter / year)

Energy kWh for the selected range, priced at the fixed tariff. Sparkline follows energy, not a separate cost series.

Formula: $\text{SAR} = \text{kWh} * 0.20$.

Source: SQLite kWh * 0.20 SAR/kWh.

Visitors today / month to date / this quarter / this year

Customer visits = door entries, never camera "events". MTD is 1st of this Riyadh month through today, including live today.

Quarter and year-to-date use the same entry counts.

Formula: $\text{visits} = \text{sum of daily entries in range}$.

Source: LC Logic /api/daily?days=365 plus /api/today.entries overlaid on today.

Visitor deltas (MTD only)

Percent vs last month, same calendar days. Percent vs same month last year. Up is treated as good (invert).

Source: LC Logic daily entries.

Visitors today so far

Always today's live entries, ignoring the range pill. Compared with the same weekday last week. Peak hour is the Riyadh hour with the most entries so far (API hours are UTC, shifted +3).

Formula: $\text{vs last week \%} = (\text{today} - \text{last weekday}) / \text{last weekday} * 100$.

Source: LC Logic /api/today.entries, /api/hourly, /api/daily for today-minus-7.

kWh per customer

Selected-range energy kWh divided by selected-range visitor entries. Sparkline is cumulative MTD kWh / cumulative MTD visits by day. "Need customer visits" if visits are 0.

Formula: $\text{kWh/customer} = \text{kWh} / \text{visits}$.

Source: SQLite closed kWh / LC Logic entries.

Saved by the system

How much cheaper this period is versus last year, same days, at 0.20 SAR/kWh. Positive means we used less than last year. Year-to-date line sums (last-year month - this-year month) * 0.20 for each month in the 12-month chart.

Formula: $\text{SAR saved} = (\text{last year kWh} - \text{this period kWh}) * 0.20$.

Source: SQLite year-ago kWh vs this period kWh.

Average dwell time

Placeholder. The visitor API does not send dwell. Always shown as a dash.

Source: Not connected.

Table use now

Share of dining table zones that currently have at least one person. People on the floor is the camera total (includes all zones).

Formula: $\text{table use \%} = \text{occupied tables} / \text{table zones} * 100$.

Source: LC Logic /api/counts, entrance zones removed.

All branches**Heading: All branches**

Chain table. Al Malqa, Riyadh is the only live row. Hittin, Al Rawdah (Jeddah), Al Faisaliyah (Dammam), Al Olaya, and Al Nakheel show "Not integrated yet" with blank metrics. Region filter: All / Riyadh / Jeddah / East.

Source: Live row uses the same Overview energy + visitors as Al Malqa. Other rows are a static list.

kWh / customer

This month's closed energy kWh divided by this month's visitor entries (MTD, not the range pill).

Source: SQLite MTD / LC Logic MTD.

Energy MTD (kWh)

Closed-day kWh from the 1st of this month through the last 6 AM close.

Source: [SQLite](#).

Saved MTD

SAR saved versus last year, same days (not versus last month). Shown as zero if we used more than last year.

Formula: $\max(0, (\text{last year same days kWh} - \text{MTD kWh}) * 0.20)$.

Source: [SQLite](#).

vs Aug (column label)

The header is leftover mockup text. The number is percent change versus last calendar month, same closed days. Red/up = using more energy.

Source: [SQLite month.changePct](#).

Trend

Sparkline of the last 14 calendar days of closed energy totals (today's open day is usually 0).

Source: [SQLite last-14-day series](#).

Status

First Needs-attention title (Cooling, Waste, Data, Power), or OK if there are no alerts. Open goes to Branch.

Source: [Same alert list as Needs attention](#).

Angle 1 - Energy saving

This month vs last month vs last year

Running total of closed kWh by day-of-month. Teal = this month. Blue = last month. Dashed grey = same month last year. Compared on the same day numbers (day 1..through last close).

Source: [SQLite](#).

Where the change came from

Month-to-date kWh split by load: Lights, AC, Oven, Freezer, Chiller. Three bars per load: this month, last month same days, last year same days.

Source: [SQLite, devices grouped by category](#).

12 month trend vs previous year

Stacked bars of each load for the last 12 months. Current month is month-to-date (closed days only). Dashed line is the same month last year.

Source: [SQLite](#).

Angle 2 - Business insights

This month vs last month vs last year (visits)

Same shape as the energy cumulative chart, but using door entries. Includes live today.

Source: [LC Logic daily entries](#).

Customer visits by daypart

Typical mix from the last 4 weeks of weekday-hour averages, then scaled to this month's calendar day number. Breakfast 6-9, Lunch 10-13, Afternoon 14-16, Dinner 17-20, Late night 21-5 (Riyadh hours).

Formula: $\text{daypart} = \text{sum}(\text{hourly average}) * \text{throughDay}$.

Source: [LC Logic /api/hourly-by-dow](#) for 7 weekdays, last 28 days. Hours shifted UTC+3.

When customers come (heatmap)

Average entries by weekday and hour, last 4 weeks. Darker = busier. Busiest / quietest look only at 10:00-23:00.

Source: [LC Logic /api/hourly-by-dow?days=28](#).

Energy by load - Closed days

Lights / AC / Oven / Freezer / Chiller cards

Last closed day's kWh for that load, plus the last 7 closed days and the 7 days before that.

Source: [SQLite](#).

Last 14 days stacked bars

Each bar is one energy day, stacked by load. Today's open day is usually empty until 6 AM tomorrow.

Source: [SQLite](#).

2. Branch

Route: /branch. Live floor view for Al Malqa. Employees see only this page. Power and climate come from Home Assistant. People counts come from LC Logic. Today Energy on devices is live meter minus 6:00 AM snapshot (same formula as Reports "so far").

Live overview

In restaurant

People currently detected across all camera zones (including entrance).

Source: [LC Logic /api/counts.total](#).

Customer visits

Entries today. Same definition as Overview "Visitors today so far".

Source: [LC Logic /api/today.entries](#).

Peak hour

Riyadh hour so far with the most entries. Hint is that hour's entry count.

Source: [LC Logic /api/hourly \(UTC hours + 3\)](#).

Table occupancy chips

Same as Overview: green if the zone has people, grey if empty. Entrance zones hidden.

Source: [LC Logic /api/counts](#).

Energy summary (closed days)

Energy today / Latest closed day

If the last SQLite day is still "today", the title says Energy today. Otherwise it is the latest closed 6 AM day. This is NOT the live since-6-AM total on the device cards below.

Source: [SQLite overview.today.total](#).

Last 7 days

Sum of the last 7 closed days vs the 7 days before that. Increase / Reduction / No change.

Formula: $\text{change \%} = (\text{this week} - \text{last week}) / \text{last week} * 100$.

Source: [SQLite week compare](#).

This month

Closed-day MTD vs last month, same days.

Source: [SQLite month compare](#).

Load chips (Lights, AC, Oven, Freezer, Chiller)

That load's kWh on the latest closed day.

Source: [SQLite](#).

Climate control

Each AC card

Live Home Assistant climate entity. Now = current_temperature. Target = setpoint (writable). Humidity if HA sends it. Mode / fan / swing call HA services.

Source: [HA climate.*](#)

Climate current / power / today energy

Mapped channel: Kitchen ch1, Office ch2, Oven ch3, Sheeter ch4, Right dining ch6, Left dining ch7. Today energy = live channel kWh minus 6 AM baseline. Power below 0 is shown as 0.

Formula: $\text{today kWh} = \max(0, \text{live} - 6 \text{ AM snapshot})$.

Source: [HA sensor.ac_energy_monitor_energy1_chN_*](#) and SQLite baseline.

Lighting

Each light card

On/off from HA light entity. Toggle calls light.turn_on / turn_off. Metrics from the matching smart breaker: current, power (floored at 0), today energy, temperature, voltage.

Source: HA light.* plus sensor.*_current / _power / _energy / _temperature / _voltage. Today energy uses 6 AM baseline.

Live cameras

Camera tiles

MJPEG (or snapshot) proxied from Home Assistant. Count in the heading is how many camera.* entities HA reports.

Source: HA camera.* via /api/camera/{entity}.

Daily Power / Energy usage

Lights Energy

Total Power = sum of the five light-breaker power sensors (negatives to 0). Today Energy = sum of each breaker's live kWh minus its 6 AM baseline. Current = sum of amps. Voltage and Temp = averages of the five breakers.

Source: HA smart_circuit_breaker.* and smart_energy_breaker, plus SQLite baselines.

AC Energy

Total Power = sum of ch1-ch7 power (negatives to 0). Today Energy prefers the HA total energy meter minus 6 AM baseline; if that baseline is missing, it sums the seven channels. Voltage and temp come from the AC monitor device.

Source: HA ac_energy_monitor_energy1.* and SQLite baselines.

Oven / Freezer / Chiller Energy

Live three-phase current, power (negated to 0), and voltage for legs A/B/C. Today Total Energy uses the *_total_energy meter minus 6 AM baseline (same idea as Reports).

Source: HA oven_energy_meter / freezer_energy_meter / chiller_energy_meter and SQLite baselines.

3. Reports (admin)

Route: /reports. Closed 6:00 AM days plus today's live "so far" rows. Tabs: Lights, AC, Oven, Freezer, Chiller. Admin only.

Device filter

Limits rows and totals to one meter, or All devices in that tab.

Source: Device list is the REPORT_DEVICES map (friendly name to HA entity).

Period filter

All time, Today, Last 7 days, This month, Last month, This year, a month grid, or a custom date range. Dates are energy-day keys (6 AM days), Asia/Riyadh.

Source: UI filter over SQLite day_key, plus live today when the range includes today.

Total Consumption (All time)

Sum of every closed-day kWh for the selected tab/device, plus today's live so-far if today is included. Independent of the period chips except that live today is always added to this all-time figure when today is in range.

Source: SQLite consumption + live todayConsumption.

In this period

Same sum, but only rows whose day_key sits inside the selected dates (and live today if the range covers today).

Source: Filtered SQLite + optional live rows.

Table row: closed day

One device, one energy day. Consumption is that day's kWh. Day is the snapshot date in Riyadh.

Source: SQLite. consumption = next 6 AM meter - this 6 AM meter, only if the next snapshot is the very next calendar day and the meter did not go backwards.

Table row: today so far

Live HA kWh minus today's 6 AM baseline. If the meter has not updated since before 6 AM, it is tagged stale and today stays 0.00 until HA sends a new total. AC meters usually keep moving; some light meters freeze overnight.

Formula: *today so far* = $\max(0, \text{live meter} - 6 \text{ AM snapshot})$.

Source: HA state + SQLite baseline.

Table footer Total consumption

Sum of consumption on the filtered rows (period total if a period is set, otherwise all-time including live).

Source: Same as the cards above.

What is stored at 6:00 AM

Lights: Dining, Counter, Kitchen, Office, Oven area breakers. AC: Kitchen, Office, Oven, Sheeter, Right dining, Left dining channels, plus a total meter and channel 5 for baselines. Oven / Freezer / Chiller: total_energy meters, plus extra energy sensors for Branch live math. Cron on the live server: 0 6 * * * with CRON_TZ=Asia/Riyadh.

4. Statistics (admin)

Route: /statistics. Visitor analytics from LC Logic only. No energy.

Entries

Today's door entries. Hidden when a weekday-average filter is selected (only shown for Today).

Source: [LC Logic /api/today.entries](#).

Peak hour

Hour with the most entries in the chart currently shown (today actual, or a weekday 30-day average).

Source: Today: [/api/hourly](#). Weekday: [/api/hourly-by-dow?days=30](#). Hours shifted UTC+3 to Riyadh.

Customers entered - by hour

Bar chart. Dropdown: Today (actual) or Sunday-Saturday average over the last 30 days. Average caption shows how many that weekday occurred in the window.

Source: [LC Logic hourly / hourly-by-dow](#). Visits in those APIs are mapped from events, but this chart uses entries.

Customers entered - last 14 days

One bar per calendar day of entries.

Source: [LC Logic /api/daily?days=14](#).

Zone totals

People currently in each named camera zone, including entrance.

Source: [LC Logic /api/counts](#).

Event log

Recent entry / exit / visit events with time, zones, camera, and event id. Rows without zones are hidden.

Source: [LC Logic /api/events?limit=200](#).

5. Schedules (admin)

Route: /schedules. Home Assistant automations. Not calculated metrics.

Each automation card

Name is HA friendly_name. Toggle turns the automation on or off. Last triggered is HA last_triggered. Run now calls automation.trigger.

Source: [HA entities whose id starts with automation.*](#)

6. Login

COMMAND DECK

Username / password gate. Admins go to Overview. Invalid credentials stay on this screen. Theme defaults to dark.

Source: [App gate \(SITE_PASSWORD / stored users\)](#). No metrics.

7. Device map (energy)

Friendly names as shown on Reports and Overview load charts. Today so far always subtracts the 6:00 AM snapshot of the same entity.

Lights

- Dining Lights - sensor.smart_energy_breaker_energy
- Counter Area - sensor.smart_circuit_breaker_counter_lights_energy
- Kitchen Area - sensor.smart_circuit_breaker_kitchen_lights_energy
- Office Area - sensor.smart_circuit_breaker_office_lights_energy
- Oven Area - sensor.smart_circuit_breaker_oven_lights_energy

AC

- Kitchen Area - ch1 energy (climate.kitchen_area)
- Office Area - ch2 energy (climate.office_area)
- Oven Area - ch3 energy (climate.oven_area)
- Sheeter Area - ch4 energy (climate.sheeter_area)
- Right Dining Area - ch6 energy (climate.dining_area_right)
- Left Dining Area - ch7 energy (climate.dining_area_left)
- AC Total (baseline / Branch card) - sensor.ac_energy_monitor_energy1_energy_total

Kitchen plant

- Oven - sensor.oven_energy_meter_total_energy
- Freezer - sensor.freezer_energy_meter_total_energy
- Chiller - sensor.chiller_energy_meter_total_energy

Quick reference

Customer visits = entries. Occupancy = people now. Table use = share of non-entrance zones with people. Closed kWh = 6 AM to 6 AM. Live today kWh = now minus 6 AM. Cost = kWh * 0.20 SAR. Overview energy excludes the open day. Overview visitors include today. Dwell time is not in the feed. Unintegrated branches have no numbers.